Uncertainty-Aware Entity-Anchored Graph Mining for Weakly Aligned Cognitive Risk Stratification

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Abstract—**Early cognitive assessment is better viewed as high-recall risk stratification than as final clinical diagnosis. Brain MRI and gut microbiome profiles are promising but weakly aligned modalities: they differ in biological scale, acquisition mechanism, temporal dynamics, and noise sources, and they rarely provide stable one-to-one feature correspondence. This paper studies weakly aligned entity-anchored multimodal graph mining for cognitive risk screening. We represent demographic and brain ROI variables as patient-node attributes and promote microbiome species from flat tabular columns to entity nodes in a patient–species heterogeneous graph. To reduce unstable cross-modal propagation, we use a variational graph information bottleneck (VGIB) with a focal risk objective. Across 454 subjects, the primary VGIB-HGNN-Focal configuration achieves a strong screening operating point (Risk Recall 0.843, Risk F1 0.657), while weak-alignment diagnostics show negative cross-modal predictability, low canonical correlations, and extremely small cross-modal neighborhood overlap. Additional analyses reveal an important nuance: microbiome graphs are not uniformly dominant over strong brain-demographic and late-fusion baselines, but they provide complementary risk evidence in graph-specific boundary subgroups. In the graph-only high-risk subgroup, VGIB improves Risk F1 over a brain-demographic logistic baseline by +0.233 and Risk Recall by +0.545. Nested β selection further confirms that bottleneck strength should be selected within validation folds rather than fixed globally. These findings suggest that entity-anchored graph mining is most credible under weak alignment when interpreted as uncertainty-aware complementary risk evidence rather than as a universal replacement for strong patient-level predictors.**

Index Terms—**heterogeneous graph neural networks, multimodal data mining, graph information bottleneck, weak alignment, risk stratification, microbiome, brain MRI**

I. Introduction

Dementia is a major public health challenge, and recent evidence suggests that a substantial fraction of dementia cases may be preventable or delayed through earlier risk-factor intervention [1]. Mild cognitive impairment (MCI) is widely treated as an important transitional stage between normal aging and dementia. In early assessment, the objective is often not a final diagnostic decision. Instead, a screening system should identify potentially high-risk individuals for follow-up before irreversible neurodegenerative damage occurs. This makes the task naturally screening-oriented: missing truly high-risk subjects is often more costly than flagging some ambiguous subjects for further assessment.

Multimodal biomedical learning is attractive in this setting. Brain MRI captures central structural degeneration, especially in regions such as the hippocampus, amygdala, and entorhinal cortex [8]–[10]. Gut microbiome profiles may reflect peripheral immune, inflammatory, and metabolic states associated with the microbiota–gut–brain axis [2], [4], [5]. These two sources appear complementary: MRI provides relatively direct neuroanatomical information, while microbiome profiles may contain peripheral biological context. However, this complementarity does not imply strong alignment. MRI reflects accumulated structural change in the central nervous system, whereas microbiome abundance is affected by diet, medication, lifestyle, environment, sequencing variation, and cohort-specific conditions. Direct early fusion may therefore introduce unstable cross-modal correlations, and alignment-based latent methods may be inappropriate when the modalities do not share a strong common subspace [13]–[15].

This work studies the resulting challenge as *weakly aligned entity-anchored multimodal graph mining*. The key idea is not to force every modality into one flat feature table. Instead, a peripheral modality can be transformed into entity anchors that connect subjects through shared biomedical entities. In our application, microbiome species become explicit nodes, while patient nodes carry demographic and brain ROI attributes. This creates a patient–species heterogeneous graph that preserves the relational nature of microbiome profiles without requiring one-to-one alignment between brain features and microbial taxa.

A central lesson from our experiments is that weakly aligned graph information should be interpreted carefully. A microbiome graph can expose complementary risk patterns, especially in boundary cases, but it is not automatically superior to strong brain-demographic predictors. We therefore frame the proposed VGIB-HGNN-Focal model as an uncertainty-aware graph branch that can provide complementary evidence under weak alignment. The model combines heterogeneous message passing, variational graph information bottleneck regularization, and focal loss. We additionally evaluate auxiliary and gated fusion diagnostics to test whether graph evidence is better used as a supporting branch rather than as the sole predictor.

Our contributions are summarized in three points:

- We formulate cognitive risk screening as a weakly aligned entity-anchored multimodal graph mining problem and empirically characterize the MRI–microbiome mismatch using reconstruction, CCA, neighborhood-overlap, and risk-marker discordance criteria.
- We develop an uncertainty-aware patient–species graph framework that combines heterogeneous message passing, variational graph information bottleneck regularization, focal risk learning, validation-selected bottleneck strength, and auxiliary fusion diagnostics.

- We provide a comprehensive evaluation on a real cohort and controlled benchmarks, showing both the value and the limits of graph evidence: VGIB-HGNN yields a strong primary screening operating point and is most informative for graph-specific boundary subgroups, while strong tabular predictors remain competitive.

II. Related Work

A. Multimodal Learning under Weak Alignment

Multimodal learning has been widely used to integrate imaging, clinical, molecular, and behavioral measurements. Standard early fusion concatenates modalities, while correlation-based or deep alignment methods seek shared representations [13]. Multimodal learning has also been studied in neuroimaging and Alzheimer's disease prediction, where imaging and non-imaging variables are combined to improve risk estimation [11], [12]. These strategies are effective when modalities provide synchronized or semantically corresponding views. In biomedical settings, however, modalities often differ in biological scale, acquisition process, measurement noise, and temporal dynamics. Survey work on multimodal learning identifies representation, alignment, and fusion as fundamental challenges under heterogeneous and incomplete modalities [14]. Work on unaligned multimodal sequences further shows that direct synchronization is not always available [15]. Our setting is more severe than temporal misalignment: brain MRI and microbiome profiles reflect different biological systems and are only weakly connected through indirect physiological pathways.

B. Brain MRI, Microbiome, and Cognitive Risk

Structural MRI has long been used to study MCI and Alzheimer's disease. Medial temporal regions such as the hippocampus and entorhinal cortex are associated with cognitive impairment and disease progression [8]–[10]. In parallel, gut microbiome alterations have been reported in Alzheimer's disease and may be related to inflammatory and neurobiological mechanisms [2]–[5]. Microbiome profiles are also compositional, sparse, and sensitive to preprocessing choices, motivating cautious use of abundance features and transformations [6], [7]. Nevertheless, microbiome measurements are sensitive to lifestyle and cohort conditions. This motivates a cautious modeling goal: microbiome information should not be treated as a standalone diagnostic biomarker, but as a potentially complementary screening signal.

C. Graph Neural Networks and Information Bottleneck

Graph neural networks (GNNs) provide a natural framework for modeling relational biomedical data [16]–[20]. Rather than assuming subjects are independent samples, graph models propagate information through patient similarities, biological entities, or disease relations. However, graph propagation may amplify noise when edges are weakly related to the target. The information bottleneck principle encourages representations that preserve task-relevant information while compressing irrelevant variation [21], while variational formulations make stochastic bottleneck learning practical in neural networks [22], [23]. Graph information bottleneck extends this perspective to graph-structured learning [24]. Since cognitive risk screening is class-imbalanced and operating-point dependent, we also report precision-recall and risk-oriented metrics rather than accuracy alone [26], [27]. We use this principle not as a generic regularizer, but as a mechanism for controlling unstable information flow under weak multimodal alignment.

III. Problem Formulation

A. Weakly Aligned Entity-Anchored Data

Let $\mathcal{D} = \{(x_i^a, X_i^e, y_i)\}_{i=1}^N$ denote a multimodal dataset. The attribute modality x_i^a contains patient-level variables, such as demographics and brain ROI features. The entity modality X_i^e measures subject–entity relations over a biomedical entity set $\mathcal{E} = \{e_1, \ldots, e_S\}$, such as microbiome species. The binary label $y_i \in \{0, 1\}$ indicates high-risk status. We call the modalities weakly aligned when they may contain complementary information but do not satisfy stable feature-level, temporal, or latent-space correspondence. In such cases, forcing all modalities into a shared representation can introduce spurious cross-modal correlations.

B. Patient–Species Heterogeneous Graph

We construct a heterogeneous graph

$$G = (\mathcal{V}_p, \mathcal{V}_s, \mathcal{E}_{ps}), \tag{1}$$

where $\mathcal{V}_p$ contains patient nodes, $\mathcal{V}_s$ contains species nodes, and $\mathcal{E}_{ps}$ contains patient–species edges. Each patient node is initialized as

$$x_i^p = [x_i^{demo} \| x_i^{brain}], \tag{2}$$

where x_i^{demo} contains demographic covariates and x_i^{brain} contains brain ROI features. For patient i and species j, the edge attribute is

$$a_{ij} = \log(1 + \mathrm{abundance}_{ij}). \tag{3}$$

This promotes microbiome species from flat features to relational anchors. The graph is not assumed to be perfectly causal or universally predictive; rather, it provides a structured representation through which peripheral biological similarity can be tested.

C. Weak-Alignment Criteria

We characterize weak alignment using four criteria. First, cross-modal predictability measures whether one modality reconstructs the other:

$$R^2_{M \to B} = 1 - \frac{\sum_i \|B_i - f(M_i)\|_2^2}{\sum_i \|B_i - \bar{B}\|_2^2}, \tag{4}$$

and analogously $R^2_{B \to M}$. Second, canonical correlation analysis (CCA) evaluates the strongest shared linear subspace:

$$\rho = \max_{u,v} \mathrm{corr}(Bu, Mv). \tag{5}$$

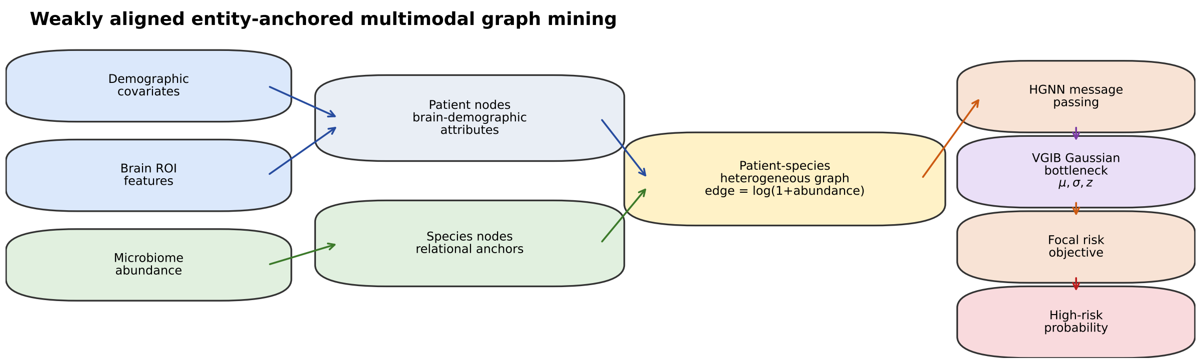

Fig. 1. Overview of the proposed weakly aligned entity-anchored graph mining framework. Patient nodes carry demographic and brain ROI attributes, while microbiome species are represented as relational anchors. A heterogeneous graph encoder propagates patient–species information, and a variational graph information bottleneck regularizes patient embeddings before focal risk prediction.

Third, neighborhood overlap compares patient topology across modalities:

$$\text{Overlap@}k = \frac{1}{Nk}\sum_i |\mathcal{N}_B^k(i) \cap \mathcal{N}_M^k(i)|. \tag{6}$$

Finally, risk-marker discordance compares top-risk sets derived from brain-demographic and microbiome-related scores:

$$\text{Jaccard}_q = \frac{|\mathcal{H}_B^q \cap \mathcal{H}_M^q|}{|\mathcal{H}_B^q \cup \mathcal{H}_M^q|}. \tag{7}$$

Together, these diagnostics distinguish weak complementarity from strong alignment.

D. Compact Notation

Table I summarizes the main symbols used in the method. The table is intentionally compact and only includes notation needed for graph construction, bottleneck learning, and screening evaluation.

TABLE I
COMPACT NOTATION USED IN THIS PAPER.

Symbol	Description
N	Number of subjects
S	Number of selected microbiome species
$G = (\mathcal{V}_p, \mathcal{V}_s, \mathcal{E}_{ps})$	Patient–species heterogeneous graph
x_i^{demo}, x_i^{brain}	Demographic and brain ROI features of patient i
M_{ij}	Abundance of species j in patient i
a_{ij}	Edge attribute, $\log(1 + M_{ij})$
h_i	Patient embedding after HGNN message passing
z_i	Stochastic latent representation after VGIB
β	KL bottleneck coefficient
γ	Focal-loss focusing parameter

IV. Method

A. Heterogeneous Message Passing

At layer l, the heterogeneous graph encoder updates patient representations by aggregating species-neighbor messages:

$$h_i^{p,(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}_s(i)} \alpha_{ij}^{(l)} W^{(l)} h_j^{s,(l)}\right), \tag{8}$$

where $\alpha_{ij}^{(l)}$ is an attention coefficient computed from patient features, species features, and edge attributes:

$$e_{ij}^{(l)} = \text{LeakyReLU}\left(a_{att}^{\top}[W_p h_i^{p,(l)} \| W_s h_j^{s,(l)} \| W_e a_{ij}]\right), \tag{9}$$

$$\alpha_{ij}^{(l)} = \frac{\exp(e_{ij}^{(l)})}{\sum_{r\in\mathcal{N}_s(i)} \exp(e_{ir}^{(l)})}. \tag{10}$$

Here a_{att} is a learnable attention vector, W_p, W_s, and W_e are learnable projections for patient, species, and edge-attribute inputs, and $\|$ denotes concatenation. This design allows microbial species to act as population-level anchors while preserving patient-level brain-demographic information.

B. Variational Graph Information Bottleneck

Let h_i denote the patient embedding. VGIB maps h_i into a Gaussian posterior:

$$\mu_i = f_\mu(h_i), \qquad \log\sigma_i^2 = f_\sigma(h_i), \tag{11}$$

$$q_\phi(z_i|h_i) = \mathcal{N}(\mu_i, \text{diag}(\sigma_i^2)), \qquad z_i = \mu_i + \sigma_i \odot \epsilon. \tag{12}$$

The posterior is regularized toward $p(z) = \mathcal{N}(0, I)$ using

$$\mathcal{L}_{KL} = -\frac{1}{2}\sum_{i,k}(1 + \log\sigma_{ik}^2 - \mu_{ik}^2 - \sigma_{ik}^2). \tag{13}$$

This stochastic bottleneck discourages the model from encoding unstable graph variation that arises from weak alignment.

C. Risk-Sensitive and Auxiliary Fusion Objectives

The primary prediction is the high-risk probability

$$\hat{p}_i = P(y_i = 1 \mid z_i). \tag{14}$$

For each sample, the probability of the ground-truth class is defined as

$$p_{t,i} = \begin{cases} \hat{p}_i, & y_i = 1, \\ 1 - \hat{p}_i, & y_i = 0. \end{cases} \tag{15}$$

Given a mini-batch or training-node set $\mathcal{B}$, the focal objective is

$$\mathcal{L}_{Focal} = -\frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \alpha_{t,i}(1 - p_{t,i})^\gamma \log(p_{t,i}), \tag{16}$$

where $\alpha_{t,i}$ balances class frequency and γ reduces the contribution of easily classified samples. The final VGIB-HGNN-Focal objective is

$$\mathcal{L} = \mathcal{L}_{Focal} + \beta \mathcal{L}_{KL}, \tag{17}$$

where β controls bottleneck strength. Because later diagnostics show that graph evidence is complementary rather than uniformly dominant, we also evaluate auxiliary fusion variants. Given a brain-demographic branch probability p_b and graph-branch probability p_g, validation-weighted fusion uses

$$p_{fused} = \lambda p_b + (1 - \lambda) p_g, \tag{18}$$

where λ is chosen inside the training fold. Learned gating uses meta-features such as p_b, p_g, their disagreement, and entropy proxies to estimate $p_{fused} = g(x)p_b + (1 - g(x))p_g$. These variants are diagnostic tools for testing whether graph evidence is more useful as an auxiliary branch under weak alignment.

D. Complexity and Scalability

Let N be the number of patients, S the number of selected species, E the number of patient–species edges, h the hidden dimension, z the latent dimension, L the number of graph layers, A the number of attention heads, and C the number of classes. Table II summarizes the computational modules. The dominant term is HGNN message passing, $O(LAEh)$, while the Gaussian bottleneck and classifier are lightweight at the present cohort scale. With $N = 454$, $S = 50$, and $E \approx$ 18,412, fold-wise graph construction, nested β selection, and out-of-fold evaluation remain feasible on a single GPU or modern CPU runtime.

TABLE II

MODULE-WISE COMPUTATIONAL COMPLEXITY.

Module	Complexity	Instantiated scale
Feature preprocessing	$O(Nd_p + NS)$	$N = 454$, $S = 50$
Graph construction	$O(NS)$	$E \approx$ 18,412 after filtering
HGNN message passing	$O(LAEh)$	$L = 2$, $A = 2$, $h = 64$
Gaussian projection	$O(Nhz)$	$h = 64$, $z = 32$
KL regularization	$O(Nz)$	$N = 454$, $z = 32$
Final classification	$O(NzC)$	$C = 2$

Algorithm 1 Training Procedure of VGIB-HGNN-Focal

Require: X^{demo}, X^{brain}, microbiome matrix M, labels y, focal parameter γ, candidate bottleneck set $\mathcal{B}$
Ensure: Trained model and out-of-fold risk probabilities $\hat{p}$

```
for each outer training/test fold do
    Fit imputation, scaling, prevalence filtering, and species selection using training subjects only.
    Construct patient features X^p ← [X^demo ‖ X^brain].
    Initialize species features X^s as trainable species embeddings.
    for each training patient i and selected species j do
        if M_ij > 0 then
            Create edge (v_i^p, v_j^s) with attribute a_ij ← log(1 + M_ij).
        end if
    end for
    if nested bottleneck selection is enabled then
        Split the training fold into inner train/validation subsets.
        Select β ∈ B by validation Risk F1 under the screening objective.
    end if
    Initialize HGNN encoder θ, Gaussian projection layers φ, and classifier ψ.
    for each training epoch do
        Encode patients: h_i ← HGNN_θ(G, X^p, X^s, a_ij).
        Compute μ_i ← f_μ(h_i) and log σ_i^2 ← f_σ(h_i).
        Sample ε ~ N(0, I) and reparameterize z_i ← μ_i + σ_i ⊙ ε.
        Predict risk probability p̂_i ← Classifier_ψ(z_i).
        Compute L_Focal over training nodes and L_KL = D_KL(q_φ(z_i|h_i) ‖ N(0, I)).
        Update θ, φ, ψ by minimizing L = L_Focal + βL_KL.
    end for
    Select the decision threshold using validation predictions only.
    Insert held-out patients through their selected species edges and output p̂.
end for
return Out-of-fold risk probabilities and trained fold-specific models.
```

V. EXPERIMENTAL SETUP

A. Dataset and Protocol

We evaluate on 454 subjects: 346 Normal, 86 MCI, and 22 Dementia. The binary task merges MCI and Dementia into a high-risk group ($n = 108$). Inputs include demographic covariates, brain ROI features, and species-level microbiome abundances. To reduce leakage, missing-value imputation, feature scaling, species filtering, species selection, graph construction, β selection, and threshold selection are performed within training folds whenever they affect model fitting or operating-point selection. We report out-of-fold predictions whenever possible.

B. Baselines and Metrics

We compare traditional models (logistic regression, random forest, XGBoost, LightGBM), neural tabular models (MLP-Focal), graph baselines (HGNN-CE and HGNN-Gaussian-Focal), VGIB-HGNN-Focal, late-fusion baselines, validation-weighted fusion, learned gated fusion, and nested-β VGIB. Metrics include AUC, PR-AUC, Macro-F1, Risk Recall, Risk Precision, and Risk F1. Since this is a first-stage screening problem, Risk Recall and Risk F1 are emphasized as operating-point metrics, while AUC and PR-AUC are reported as ranking metrics.

TABLE III
RESEARCH QUESTIONS AND EVIDENCE.

RQ	Purpose	Evidence
RQ1	Screening performance	Primary CV metrics
RQ2	Weak alignment	R^2, CCA, overlap, discordance
RQ3	Bottleneck, ablation, fusion	Sensitivity, ablation, fusion diagnostics
RQ4	Controlled relational benchmark	Synthetic graph-position recovery
RQ5	Deployment diagnostic	Inductive patient insertion
RQ6	Boundary cases and interpretability	Subgroups, profiles, t-SNE

C. Research Questions

Table III summarizes the six research questions. The goal is not only to report a single performance table, but to test whether weak alignment exists, whether bottleneck regularization is necessary, whether graph evidence can be used as auxiliary information, whether relational signals are recoverable, whether the model supports inductive patient insertion, and where graph information contributes in the real cohort.

D. Implementation and Reproducibility Notes

All reported real-cohort metrics are computed from out-of-fold predictions. For tabular baselines, preprocessing is fitted within each training fold and applied to the held-out fold. For graph models, the selected species set and patient–species graph are reconstructed inside each fold to avoid using held-out distributional information for feature selection. Operating thresholds are selected only from training or validation predictions. This is important because risk-screening metrics such as Risk Recall and Risk F1 can be overly optimistic if thresholds are chosen after observing test labels.

For the graph branch, patient nodes are initialized with demographic and brain ROI features, while species nodes use learned embeddings. Edge attributes use log-transformed abundance unless otherwise stated. The focal parameter is fixed in the main experiments, while the bottleneck coefficient is either specified by the primary configuration or selected using nested validation. The auxiliary fusion diagnostics use only out-of-fold branch probabilities and simple meta-features derived from prediction disagreement, which avoids training the gate directly on held-out labels. These implementation choices are designed to keep the evaluation focused on whether graph evidence provides additional screening information under weak alignment.

We release the experimental outputs as separate tables for primary performance, weak-alignment diagnostics, bottleneck sensitivity, nested β selection, auxiliary fusion, inductive insertion, and subgroup analysis. This separation is intentional: results from different validation protocols are not merged into a single leaderboard. Instead, each table answers a distinct methodological question. This organization reduces the risk of overclaiming and makes the evidence chain easier to audit.

VI. RESULTS

A. RQ1: Primary Screening Performance

Table IV reports the primary screening results. VGIB-HGNN-Focal achieves the strongest screening-oriented operating point in the primary experiment suite, with Risk Recall 0.843 and Risk F1 0.657. The model does not dominate every ranking metric; HGNN-Gaussian-Focal has slightly higher AUC/PR-AUC. The intended claim is therefore not universal dominance, but a favorable high-risk screening trade-off under the primary configuration.

TABLE IV
PRIMARY SCREENING PERFORMANCE.

Model	AUC	PR-AUC	Recall	Precision	F1
LogReg-BrainDemo	0.858	0.701	0.759	0.516	0.614
LogReg-AllTabular	0.792	0.579	0.661	0.487	0.560
RF-BrainDemo	0.832	0.697	0.776	0.496	0.604
RF-AllTabular	0.849	0.727	0.778	0.506	0.613
MLP-Focal-BrainDemo	0.844	0.728	0.806	0.483	0.604
MLP-Focal-AllTabular	0.825	0.691	0.737	0.479	0.580
HGNN-CE	0.843	0.720	0.815	0.497	0.618
HGNN-Gaussian-Focal	0.859	0.735	0.833	0.511	0.634
VGIB-HGNN-Focal	0.853	0.733	**0.843**	**0.538**	**0.657**

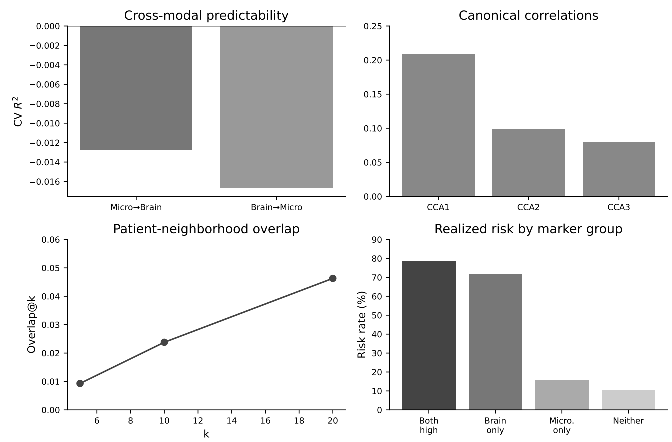

Fig. 2. Weak-alignment diagnostics.

B. RQ2: Weak Alignment is Empirically Strong

Figure 2 summarizes the weak-alignment evidence. Cross-modal predictability is negative in both directions ($R^2_{M\to B} = -0.013$, $R^2_{B\to M} = -0.017$), indicating that one modality cannot reliably reconstruct the other out of fold. The first three CCA correlations are low (0.209, 0.099, 0.080). Neighborhood overlap is also extremely small: Overlap@5 = 0.009, Overlap@10 = 0.024, and Overlap@20 = 0.046. Finally, risk-marker grouping shows discordance: the Both-High group has the highest realized risk rate, but Brain-only and Microbiome-only high-risk sets are not equivalent. This supports the central premise that microbiome information should not be forced into a strongly aligned representation with MRI.

C. RQ3: Bottleneck Sensitivity, Ablation, and Fusion

The bottleneck coefficient is not a cosmetic hyperparameter. Figure 3 shows that Risk F1 changes materially across β values. In one sensitivity run, $\beta = 5 \times 10^{-4}$ gives the

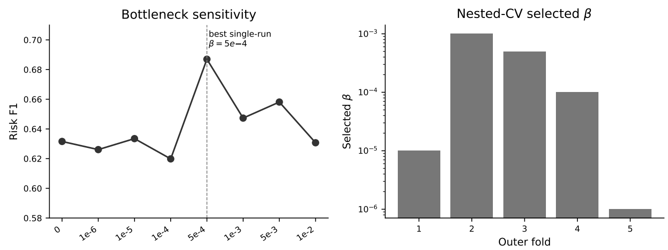

Fig. 3. Bottleneck sensitivity and nested β selection.

TABLE V
AUXILIARY FUSION AND STRONG-BASELINE DIAGNOSTICS.

Model	AUC	PR-AUC	Recall	Precision	F1
XGBoost-AllTabLog	0.852	0.675	0.685	0.649	**0.667**
XGBoost-BrainDemo	0.848	0.719	0.611	**0.733**	**0.667**
HGNN-CE	0.843	0.726	0.648	0.686	**0.667**
HGNN-Gaussian-Focal	0.853	0.727	**0.704**	0.608	0.652
VGIB-HGNN-Focal	0.849	**0.737**	0.657	0.602	0.628
Validation-weighted	0.855	0.714	0.667	0.600	0.632
Uncertainty-gated	**0.858**	0.717	0.676	0.589	0.629
Nested-β VGIB	0.827	0.706	0.694	0.605	0.647

strongest Risk F1 (0.687). However, to avoid post-hoc test-set tuning, we also conduct nested validation: each outer fold selects β using inner validation. The selected values vary across folds (10^{-5}, 10^{-3}, 5×10^{-4}, 10^{-4}, and 10^{-6}), and the nested-β model retains competitive screening performance (Risk Recall 0.694, Risk F1 0.647). This supports a more careful interpretation: bottleneck strength should be validation-selected because weak-alignment noise and useful graph signal vary across folds.

1) Auxiliary Fusion Diagnostics: Table V compares stronger baselines and auxiliary fusion variants. XGBoost and HGNN-CE are strong at the selected operating point. Learned gated fusion obtains the highest AUC (0.858), while VGIB-HGNN-Focal has the highest PR-AUC among the listed graph/fusion diagnostics (0.737). However, gated fusion does not consistently improve Risk F1 over the strongest baselines. This result is important: under weak alignment, microbiome graph evidence should be viewed as auxiliary risk evidence rather than as a universally dominant predictor.

2) Feature and Relation Ablation: We further examine feature-level and graph-relation contributions under a fixed controlled protocol. The purpose of this ablation is not to reproduce the exact threshold-selected operating point of RQ1, but to isolate how patient attributes and microbiome relations change performance under a shared setting. Demographic-only features tend to produce higher recall, while brain-only features provide higher precision. Combining demographic and brain ROI features improves ranking performance, and adding real microbiome relations increases Risk F1 from 0.597 to 0.630 in the controlled ablation. Randomized microbiome edges reduce the benefit, suggesting that graph relations can contribute beyond simple model capacity, although later robustness diagnostics also show that this contribution should be interpreted cautiously.

TABLE VI
FEATURE AND RELATION ABLATION.

Setting	AUC	PR-AUC	Recall	Precision	F1
Demographic only	0.800	0.627	0.604	0.494	0.539
Brain only	0.795	0.680	0.581	0.594	0.587
Demo + Brain	0.839	0.714	0.631	0.571	0.597
Demo + Brain + Real graph	**0.842**	**0.715**	**0.654**	**0.616**	**0.630**
Randomized graph	0.842	0.703	0.633	0.609	0.619

TABLE VII
CONTROLLED GRAPH-POSITION BENCHMARK.

Model	AUC	PR-AUC	Recall	Precision	F1	Rel.-Recall
LogReg-Local	0.492	0.261	0.902	0.241	0.378	0.886
RF-Local	0.506	0.266	0.881	0.236	0.372	0.875
MLP-EarlyFusion	0.536	0.297	0.783	0.243	0.367	0.783
HGNN-RandomEdges	0.507	0.260	0.804	0.240	0.362	0.778
HGNN-RealEdges	0.841	0.626	0.661	0.536	0.575	0.652
HGNN-Gauss-Real	**0.920**	**0.796**	0.676	**0.738**	0.683	0.656
VGIB-Tunedβ	0.916	0.781	**0.723**	0.683	**0.699**	**0.716**

D. RQ4: Controlled Relational Benchmark

Real clinical data cannot fully isolate whether graph models can recover relational signals. We therefore use a controlled second-order graph-position benchmark in which labels depend on relational graph position rather than local abundance summaries. Table VII shows that local-summary models remain near random AUC, while real-edge graph models recover the relational signal. VGIB-HGNN-Focal with tuned β achieves the best Risk F1 and relational-only recall in this controlled setting. This benchmark does not prove that all real-world microbiome signals are second-order topology. Instead, it validates the methodological claim that if weakly aligned entity-anchored risk information is relational, flat local summaries can miss it.

E. RQ5: Deployment Diagnostic: Inductive Patient Insertion

To evaluate practical use for unseen subjects, we compare the standard transductive cross-validation graph with an inductive patient-insertion setting. In the inductive setting, the model is trained on the training graph, and held-out patients are inserted at inference time through their species edges without using their labels. The inductive setting remains stable, with AUC 0.850, PR-AUC 0.724, Risk Recall 0.713, Risk Precision 0.611, and Risk F1 0.658. This result suggests that the entity-anchored formulation can be used for unseen patients and is not merely exploiting test-node access in a transductive graph.

F. RQ6: Boundary Cases and Interpretability

The clearest complementary value of graph evidence appears in graph-specific boundary cases. We define a graph-only high-risk subgroup as subjects flagged by the graph

TABLE VIII
INDUCTIVE PATIENT-INSERTION DIAGNOSTIC.

Setting	AUC	PR-AUC	Recall	Precision	F1
Transductive CV	0.841	0.720	0.694	0.560	0.620
Inductive insertion	**0.850**	**0.724**	**0.713**	**0.611**	**0.658**

TABLE IX
GRAPH-ONLY SUBGROUP GAINS.

Model	Δ F1	Δ Recall	Δ AUC
VGIB-HGNN-Focal	**+0.233**	**+0.545**	**+0.500**
Validation-weighted	+0.212	+0.455	+0.327
Nested-β VGIB	+0.191	+0.455	+0.182
Uncertainty-gated	+0.188	+0.364	+0.209
RF-gated fusion	+0.161	+0.273	+0.173

branch but not by the brain-demographic branch. This subgroup contains 21 subjects with a realized risk rate of 52.4%, much higher than the overall cohort risk rate. In this subgroup, VGIB-HGNN-Focal improves Risk F1 over the brain-demographic logistic baseline by +0.233 and Risk Recall by +0.545 (Table IX, Fig. 4). Validation-weighted fusion, nested-β VGIB, and learned gated fusion also show positive gains. This suggests that graph evidence may be most informative for ambiguous subjects whose risk is not fully captured by brain-demographic predictors.

1) Boundary Profile Plausibility: Beyond metric gains, we examine whether boundary predictions are plausible. Subjects predicted as high risk despite normal labels are not interpreted as clinical diagnoses; rather, they represent candidates whose profiles may deserve follow-up in a screening workflow. Table X shows that the Normal-to-Risk group is older, has fewer education years, and has lower MMSE scores than correctly predicted normal subjects. The same direction is observed in cognition-related brain regions: hippocampus, amygdala, and entorhinal cortex volumes are smaller in the Normal-to-Risk group, with all listed differences significant after FDR correction in the original boundary analysis.

Microbiome findings are interpreted more cautiously. Because abundance data are compositional and highly skewed, both log-transformed (LOG) and centered log-ratio (CLR) features are examined. Table XI shows that CLR-transformed *Parabacteroides* unclassified reaches FDR-corrected significance ($q = 0.0469$), while microbiome richness and selected LOG-transformed taxa show exploratory trends. These taxa-level findings should not be treated as confirmatory biomarkers. Their role is to support the screening interpretation: graph-flagged boundary subjects differ not only in model scores but also in demographic, neuroanatomical, and partially peripheral biological profiles.

2) Qualitative Embedding Visualization: We additionally inspect patient embeddings using t-SNE as qualitative evidence only. This visualization is not used as a primary performance estimate because it trains a single visualization model on the cohort. Nevertheless, it helps illustrate the

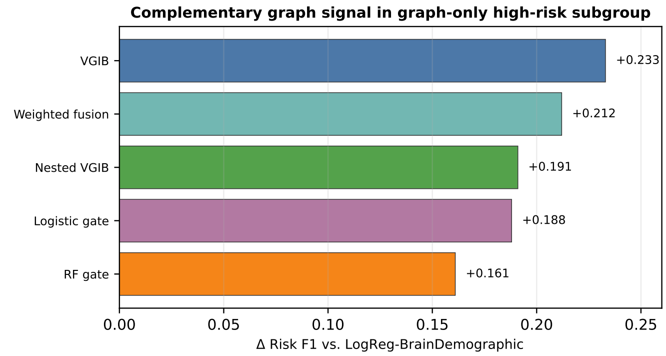

Fig. 4. Risk F1 gains in the graph-only high-risk subgroup.

TABLE X
CLINICAL AND BRAIN PROFILES OF BOUNDARY CASES.

Feature	N-to-Risk	Correct Normal	Direction	Effect	q
Age	73.220	65.330	Higher	1.325	< 0.001
Education years	10.410	13.600	Lower	-0.801	< 0.001
MMSE	26.820	28.300	Lower	-0.661	0.004
Hippocampus mean	3954.650	4422.150	Lower	-1.205	< 0.001
Amygdala mean	1520.270	1741.720	Lower	-1.243	< 0.001
Entorhinal cortex mean	1643.890	1771.780	Lower	-0.572	< 0.001

representation effect of the bottleneck. Figure 5 compares the HGNN pre-bottleneck embedding with the VGIB latent mean. The silhouette score by true risk label increases from 0.238 to 0.317, and a linear probe AUC increases from 0.903 to 0.910. These diagnostics suggest that the bottleneck slightly improves risk-label organization in the learned representation while maintaining a continuum of boundary subjects rather than a hard diagnostic separation.

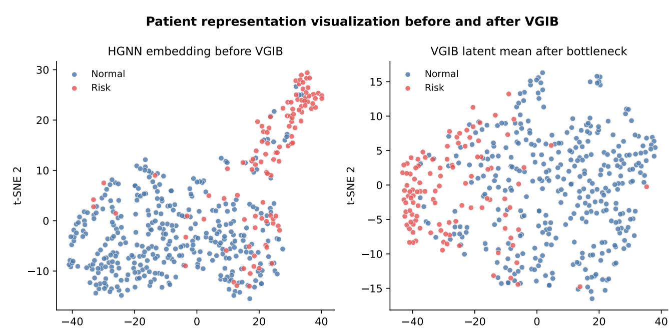

Fig. 5. Qualitative t-SNE visualization before and after VGIB.

G. Empirical Training Efficiency

Table XII reports the empirical average end-to-end training time measured on Google Colab with a T4 GPU. The proposed VGIB-HGNN-Focal model incurs additional cost from patient–species message passing and Gaussian latent inference, but remains comparable to standard HGNN variants and practical for the present medium-scale biomedical screening cohort.

TABLE XI
MICROBIOME PROFILES OF BOUNDARY CASES.

Feature	N-to-Risk	Correct Normal	Effect	q	Evidence
CLR *Parabacteroides* unclassified	-6.303	-3.529	-0.422	0.047	FDR sig.
Microbiome richness	123.040	115.820	0.411	0.103	Exploratory
LOG *Parabacteroides* unclassified	0.168	0.302	-0.257	0.103	Exploratory
LOG *Streptococcus parasanguinis*	0.413	0.239	0.402	0.103	Exploratory
LOG *Parabacteroides merdae*	0.647	0.444	0.407	0.166	Exploratory

TABLE XII
EMPIRICAL TRAINING TIME ON COLAB T4 GPU.

Model family	Mean end-to-end training time
MLP-Focal	1.1–1.4 s
Random Forest	7–8 s
Standard HGNN variants	9–14 s
VGIB-HGNN-Focal	13.4 s

VII. DISCUSSION

The experiments support a nuanced conclusion. First, weak alignment between brain MRI and microbiome profiles is empirically strong: the modalities are poorly reconstructable from each other, have weak shared linear structure, and define different patient neighborhoods. This justifies modeling design choices that avoid forcing strict cross-modal alignment. Second, entity-anchored graph modeling is useful as a representation strategy for weakly aligned peripheral signals, but it should not be overclaimed as a universal replacement for strong patient-level predictors. Strong brain-demographic and late-fusion baselines remain competitive. Third, the information bottleneck is important because useful relational signal and weak-alignment noise vary across folds; nested validation should therefore select β rather than fixing it globally. Finally, the graph branch provides its clearest clinical screening value in boundary subgroups, especially subjects identified as high risk by graph evidence but not by brain-demographic models.

This interpretation is deliberately conservative. Additional topology-control diagnostics indicate that species-specific real graph topology is not always consistently superior to randomized or perturbed graph variants in the real cohort. Therefore, our claim is not that the observed microbiome topology is universally predictive. Rather, the contribution is a framework and evidence chain for evaluating entity-anchored graph evidence under weak alignment, together with evidence that such graph evidence can reveal complementary high-risk subjects.

From a data-mining perspective, the findings emphasize that weakly aligned multimodal learning should be evaluated beyond average classification metrics. A method may not uniformly dominate the full cohort while still identifying a meaningful subgroup that a strong baseline misses. This is particularly relevant for screening tasks, where the operational value of a model may come from prioritizing ambiguous cases for follow-up rather than replacing existing predictors. The proposed framework therefore connects graph mining, uncertainty-aware representation learning, and screening-oriented subgroup analysis.

VIII. LIMITATIONS AND FUTURE OPPORTUNITIES

A. Limitations

The real cohort is modest in size, and independent external validation is needed. Microbiome measurements are sensitive to lifestyle, medication, diet, and environmental conditions, so species-level findings should not be interpreted as causal biomarkers. Some robustness diagnostics also show that graph evidence is complementary rather than uniformly dominant; accordingly, the patient–species graph should be viewed as a data-mining representation rather than a mechanistic causal explanation. Finally, threshold-based screening metrics depend on the intended operating point, and deployment thresholds should be chosen according to clinical resource constraints and follow-up capacity.

B. Future Data Mining Opportunities

This study suggests several data-mining directions. First, future work should extend static patient–species graphs into longitudinal or dynamic GNNs that model transitions from Normal to MCI and Dementia. Such models could capture whether microbiome-derived relational neighborhoods precede or follow neuroanatomical change. Second, microbiome graph semantics can be enriched beyond a bipartite abundance graph by incorporating species co-occurrence, metabolite pathways, functional annotations, and curated microbiome knowledge graphs. Third, cross-cohort transfer learning and domain adaptation are needed to handle sequencing-protocol variation and hospital-specific distribution shifts. Bayesian uncertainty and calibration-aware adaptation may be especially useful when graph evidence is weakly aligned and cohort-dependent.

IX. CONCLUSION

We presented an uncertainty-aware entity-anchored graph mining framework for cognitive risk stratification under weak multimodal alignment. The study shows that brain MRI and microbiome profiles are weakly aligned across reconstruction, latent correlation, topology, and risk-marker criteria. VGIB-HGNN-Focal provides a strong screening-oriented operating point in the primary experiment suite and, more importantly, offers complementary graph-based evidence for graph-only high-risk boundary subjects. Nested β selection and auxiliary fusion diagnostics further show that graph evidence should be regularized and used cautiously under weak alignment. These results suggest that entity-anchored graph mining is a promising but nuanced direction for multimodal biomedical screening: its value lies not in universally replacing strong patient-level predictors, but in identifying complementary risk patterns that flat fusion may miss.

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